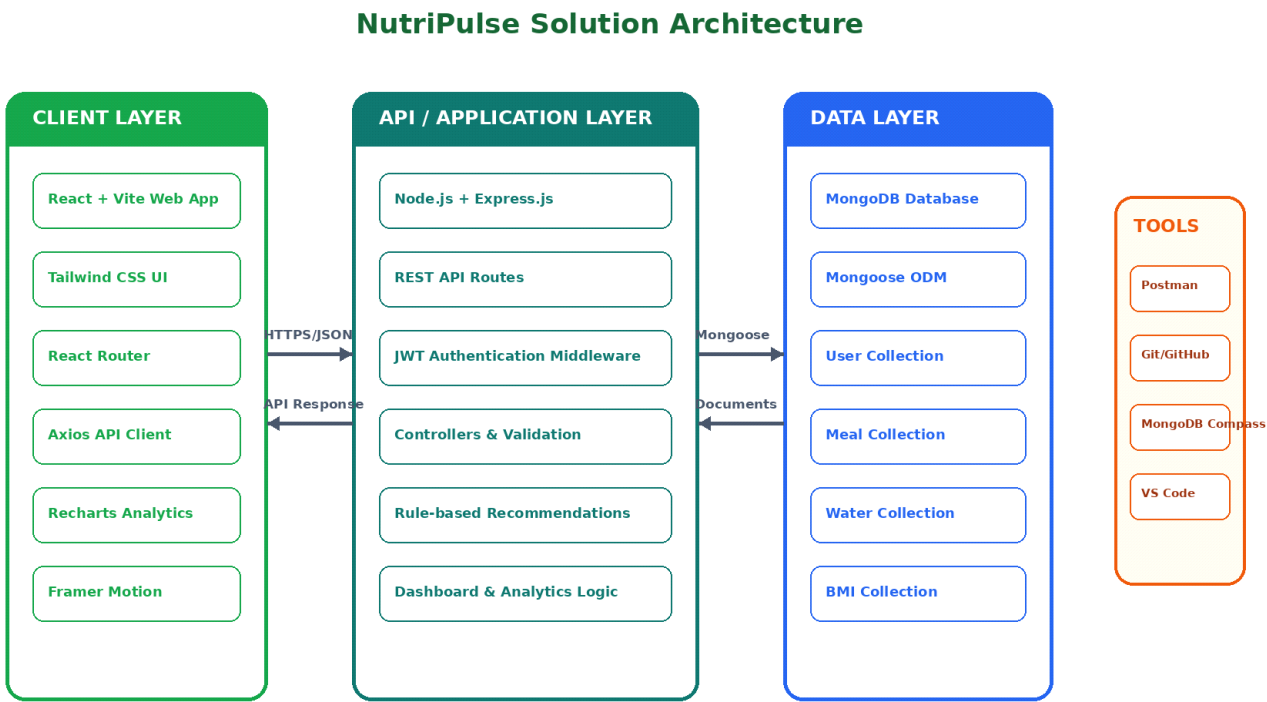


Project Design Phase

Solution Architecture

Date	19 July 2026
Team ID	
Project Name	Nutrition Assistant
Team Members	Kulasri Durga Deepika Bora Polimera Girija(Team lead)
Maximum Marks	4 Marks

Architecture Diagram



Architecture Description

Layer	Responsibilities
Client Layer	React + Vite application renders landing, authentication, dashboard, meal, water, BMI, profile and analytics pages. Axios communicates with the API and automatically sends JWT tokens.

API Layer	Express routes receive requests, apply middleware and forward operations to controllers. Controllers validate data, enforce ownership, perform calculations and return JSON responses.
Security Layer	Passwords are hashed with bcryptjs. JWT middleware validates protected requests. CORS limits allowed frontend origins and dotenv protects configuration values.
Data Layer	MongoDB stores users, meals, water entries and BMI records. Mongoose schemas provide validation, references, timestamps and query indexes.
Analytics Layer	The server aggregates daily and weekly health records, while Recharts visualizes calories, water, meals and nutrients in the frontend.
Recommendation Layer	Rule-based logic evaluates profession, goal, water, meals, calories, protein and BMI to return prioritized wellness suggestions.

Request and Data Flow

1. The user interacts with the React interface and submits registration, login or health-tracking data.
2. Axios sends a JSON request to the Express API. For protected endpoints, it includes the JWT in the Authorization header.
3. Express middleware parses the request, checks CORS and validates the JWT.
4. The selected controller validates fields and executes business logic.
5. Mongoose reads or writes the user's data in MongoDB.
6. The backend returns a structured JSON response.
7. React updates forms, cards, progress indicators, lists and charts using the response.

Major API Modules

API Module	Purpose
/api/auth	Register, login and current-user authentication
/api/profile	View and update user profile
/api/meals	Meal CRUD and nutrition totals
/api/water	Add, retrieve and delete daily water records
/api/bmi	Calculate BMI, retrieve history/latest and delete records
/api/dashboard	Return profile, today's summary, recent meals and recommendations
/api/analytics/weekly	Return seven-day totals, averages and chart data

Future Architecture Extensions

- Notification service for meal and hydration reminders
- Cloud image storage for profile photos

- Admin and reporting services
- External food nutrition API integration
- Mobile application using the same REST APIs